

Sanjay Samala

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OBJECTIVE

Software Engineer focused on building scalable applications powered by AI. I combine strong programming skills with tools like Python and modern web frameworks to transform complex Generative AI models into useful, production-ready software.

EDUCATION

Matrusri Engineering College

Bachelor of Engineering in Computer Science and Engineering

Nov. 2022 – May 2026

CGPA: 6.9

EXPERIENCE

AI Developer Intern — *Viswam.ai [Swecha Telangana, IIIT Hyd]*

June 2025 – Present

- Leading development of **large-scale Telugu language datasets** for downstream LLM training and RAG systems.
- Designed and implemented **metadata extraction pipeline** for Telugu Story Books (**758 books, 10,000+ stories, 1947-2012**) using LLM-assisted extraction with Gemini 2.5 Flash API.
- Architected **structured JSON schemas** for story-level indexing that enable precise retrieval in multilingual RAG applications.
- Built **Telugu OCR fine-tuning datasets** with **110K+ paragraph-level image-text pairs** from historical documents (1947-2012).

PROJECTS

• **Telugu Agentic RAG** — *Generative AI, Streamlit, Google Gemini OCR, RAG*

[GitHub] [Live Link]

- Designed an **Agentic RAG** architecture utilizing LangGraph, transforming **monolithic LLM generation** into a multi-stage **Plan-Draft-Critique-Polish** loop to improve output quality.
- Implemented a Self-Refining (**Actor-Critic**) workflow where a Validator Agent iteratively critiques and repairs generated content against stylistic constraints, significantly reducing hallucinations and poor translation artifacts.
- We Developed a **semantic search ingestion pipeline** using **Qdrant Vector DB** and **Sentence Transformers**, enabling style-grounded Dense Passage Retrieval based on high-dimensional vector similarity.
- Developed a **Plan-and-solve prompt optimization layer** that dynamically structures user intent into narrative blueprints prior to generation.

• **NatyaBhoda – AI Dance Learning System** — *Computer Vision, MediaPipe, XGBoost, Streamlit*

[GitHub]

- Engineered a real-time **gesture recognition pipeline** that utilizes pose estimation to classify Kuchipudi hand mudras.
- Developed a low-latency Streamlit application that **provides live inference and instant feedback** for AI-assisted dance learning.

TECHNICAL SKILLS

- **AI, GenAI & Vision:** Agentic RAG, Multi-Agent Systems (LangGraph/LangChain), Reflexion, LLMs, Dense Vector Search, MediaPipe, OCR, Prompt Engineering
- **Languages & Databases:** Python, C, JavaScript, SQL, HTML, PostgreSQL, MongoDB, Qdrant (Vector DB)
- **Frameworks & Libraries:** PyTorch, Hugging Face, Sentence Transformers (GTE), Scikit-learn, XGBoost, Pandas, NumPy
- **Backend & Web:** FastAPI, Flask, Express.js, React.js, Streamlit, REST APIs, Git

ACHIEVEMENTS AND CERTIFICATIONS

- **Ranked Top 40% (Under 1200 out of 3000 teams)** in the **Amazon ML Challenge 2024**, competing on large-scale real-world machine learning problems.
- **PHP & MySQL Certification** (4 Credits) – Spoken Tutorial Project, IIT Bombay.
- **RDBMS PostgreSQL Certification** (Score: 85%) – Spoken Tutorial Project, IIT Bombay.
- **Python Certification** (4 Credits) – Spoken Tutorial Project, IIT Bombay.